

SAMUEL COLMENARES

Katy, TX | 832-459-5598 | samucolme@gmail.com

MECHANICAL ENGINEERING STUDENT | CAD | MECHANICAL DESIGN | ENGINEERING ANALYSIS | PROTOTYPING

Texas A&M Mechanical Engineering student focused on CAD, mechanical design, solid mechanics, engineering analysis, prototyping, and practical problem-solving, with hands-on experience in SolidWorks, AutoCAD, 3D printing, assemblies, technical drawings, and experimental validation.

EDUCATION

Texas A&M University, College Station, TX

Expected 2028

Bachelor of Science in Mechanical Engineering | GPA: 3.16

Relevant Coursework: Statics; Solid Mechanics; MEEN 210 (SolidWorks/Mechanical Design); Thermodynamics (Fall 2026); Dynamics (planned before Summer 2027)

TECHNICAL SKILLS

CAD & Design: SolidWorks, AutoCAD, Autodesk Inventor, Revit; 3D modeling, assemblies, mechanical design, technical drawings, 3D printing

Engineering Analysis: Solid mechanics; bending/von Mises stress; factor of safety; design optimization; prototype testing and validation

Programming & Tools: Python; Microsoft Excel; Microsoft Project; Microsoft Word

Languages: Fluent in English and Spanish; Basic Italian

Certifications: Autodesk Certified User: AutoCAD®; Intro to AutoCAD; AutoCAD Essentials

ENGINEERING & TECHNICAL PROJECTS

MEEN 305 - Structural Beam Optimization & Experimental Validation

- Developed Design Iterations 16-20 for a beam subjected to two load cases, applying bending/von Mises stress analysis and a minimum factor of safety of 1.5 while minimizing material.
- Created CAD for Design Iteration 18, refined the Excel optimization workflow, and supported simulations and experimental test setup for 3D-printed PLA validation.
- Team prototype measured 0.00882 lb versus 0.00840 lb predicted (about 5% weight error), providing experience comparing analytical predictions with physical test results.

MEEN 210 - SolidWorks Mechanical Motion Toy Design & 3D Printing

- Created SolidWorks parts and drawings for the right arm, lower right leg, upper right leg, and right arm link in a gear- and linkage-driven mechanical motion system.
- Participated in assembly, 3D fabrication, and evaluation of fit, alignment, gear interaction, and prototype behavior during iterative design refinement.

Additional CAD Projects

- Created mechanical parts and assemblies in Autodesk Inventor, AutoCAD technical drawings, and residential architectural models in Revit.

EXPERIENCE

Funded Trader - Topstep

- Earned access to proprietary trading capital by demonstrating consistent profitability, disciplined risk management, and data-driven decision-making in simulated and live markets.
- Developed and backtested strategies across equities, futures, and forex, using structured analysis to evaluate performance and manage risk.

Bobby Sanders Basketball Camp

Finance & Operations Assistant

- Managed concession operations by tracking daily sales, handling cash flow, and maintaining accurate financial records during events.
- Provided customer service in a fast-paced environment while coordinating with staff to keep operations organized and efficient.

LEADERSHIP & COMMUNITY INVOLVEMENT

Regnum Christi

Team Leader / Formator

Present

- Lead and mentor team members in formation activities, facilitate group discussions, and support personal development through clear communication, accountability, and example.

Regnum Christi International Formators Course (IFC), Rome, Italy

Summer 2026

Participant - International Leadership & Formation Program

- Completed a 125-hour (3 ECTS) international formation program through the Pontifical Athenaeum Regina Apostolorum, with coursework in psychology, philosophy, and theology.
- Developed leadership, communication, teamwork, and cross-cultural collaboration skills through intensive formation and group activities.

Additional Service: Regnum Christi & Open Door Missions (Houston, TX); Saint Faustina Catholic Church (Katy, TX)

HONORS & AWARDS

National Hispanic Recognition Award (2022) | ATA First Degree Black Belt | Sophomore Basketball District Champions (2021-22) | Alodia Basketball Tournament Winner (2023)